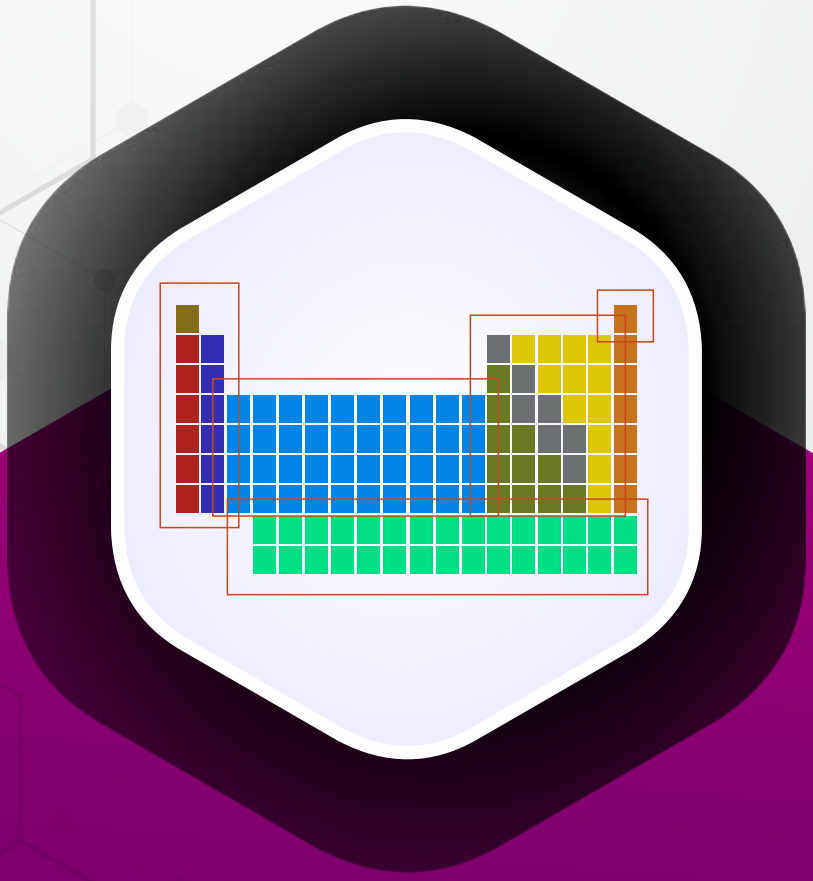
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
path to success

INORGANIC CHEMISTRY

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

Periodic Table & Periodic Properties

(English Medium)

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PART-A (PERIODIC TABLE)

INTRODUCTION :

The arrangement of all the known elements according to their properties in such a way that the elements of similar properties are grouped together in a tabular form is called periodic table.

Development of periodic table :

(A) LAVOISIER CLASSIFICATION OF ELEMENTS

At first Lavoisier classified the elements into two categories.

(i) Metal

(ii) Non-metal

Note : This classification cannot categories the metalloid variety.

(B) DOBEREINER'S TRIAD RULE [1817]

(i) He made groups of three elements having similar chemical properties called TRIAD.

(ii) In Dobereiner triad, atomic weight of middle element is nearly equal to the average atomic weight of first and third element.

e.g.

Cl	Br	I		
35.5	80.0	127	$\frac{35.5+127}{2}=81.25$	
Ca	Sr	Ba		
40	87.5	137	$\frac{40+137}{2}=88.5$	
Li	Na	K		
7	23	39	At. wt of Na $=\frac{7+39}{2}=23$	

(iii) Other examples – (K, Rb, Cs), (P, As, Sb), (S, Se, Te), (H, F, Cl), (Sc, Y, La)

(C) NEWLAND'S OCTAVE RULE [1865]

(i) He arranged the elements in the increasing order of their atomic mass and observed that properties of every 8th element was similar to the 1stone, like in the case of musical vowels notation.

Sa	Re	Ga	Ma	Pa	Dha	Ni	Sa
1	2	3	4	5	6	7	8

(ii) At that time inert gases were not known.

Li	Be	B	C	N	O	H
Na	Mg	Al	Si	P	S	F
K	Ca					Cl

(iii) The properties of Li are similar to 8th element i.e. Na, Be are similar to Mg and so on.

Drawbacks or Limitations :

(a) This rule is valid only upto Ca. because after Ca due to filling of d-orbitals there is difference of 18 elements instead of 8 element.

(b) After the discovery of inert gas this law had to be dropped out.

(c) He failed in the case of heavier metals as Fe has been placed along with O and S.

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(D) LOTHER MEYER'S CURVE [1869]

(i) He plotted a curve between atomic weight and atomic volume of different elements.

(ii) The following observations can be made from the curve –

(a) Most electropositive elements i.e. alkali metals (Li, Na, K, Rb, Cs etc.) occupy the peak portions of the curve.

(b) Less electropositive i.e. alkaline earth metals (Be, Mg, Ca, Sr, Ba) occupy the descending portions of the curve.

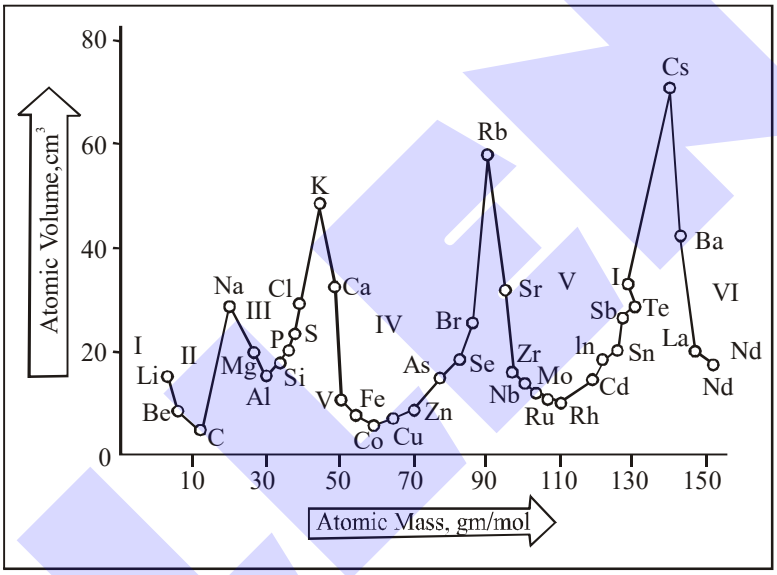
(c) Metalloids (B, Si, As, Te, At etc.) and transition metals occupy bottom part of the curve.

(d) Most electronegative i.e. halogens (F, Cl, Br, I) occupy the ascending portion of the curve.

Note : Elements having similar properties occupy similar portions of the curve.

Conclusion :

On the basis of this curve, Lother Meyer proposed that the physical properties of the elements are periodic function of their atomic weight and this became the basis of Mendeleev's periodic table.



(E) MENDELEEV'S PERIODIC TABLE [1869]

(i) ***Mendeleev's periodic law –*** The physical and chemical properties of elements are the periodic function of their atomic weight

(ii) ***Characteristic of Mendeleev's periodic table –***

(a) It is based on atomic weight

(b) 63 elements were known, noble gases were not discovered.

(c) He was the first scientist to classify the elements in a systamatic manner i.e. in horizontal rows and in vertical columns.

(d) Horizontal rows are called periods and there were 7 periods in Mendeleev's Periodic table.

(e) Vertical columns are called groups and there were 8 groups in Mendeleev's Periodic table.

(f) Each group upto VIIth is divided into A & B subgroups.'A' sub group elements are called normal elements and 'B' sub group elements are called transition elements.

(g) The VIIIth group consists of 9 elements in three rows.

(h) The elements belonging to same group exhibit similar properties.

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(iii) Merits or advantages of Mendeleev's periodic table –

(a) Study of elements – For the first time, all known elements were classified in groups according to their similar properties. So study of the properties of elements became easier.

(b) Prediction of new elements – It gave encouragement to the discovery of new elements as some gaps were left in it.

Sc (Scandium), Ga (Gallium), Ge (Germanium), Tc (Technetium)

were the elements whose position and properties were well defined by Mendeleev even before their discoveries and he left the blank spaces for them in his table.

e.g. - Blank space at atomic weight 72 in silicon group was called Eka silicon (means properties like silicon) and element *(discovered later)* was named Germanium .

Similarly other elements discovered after mendeleev periodic table were :

Eka aluminium – Gallium(Ga)

Eka Boron – Scandium (Sc)

Eka Silicon – Germanium (Ge)

Eka Manganese – Technetium (Tc)

(c) Correction of doubtful atomic weights–Corrections were done in atomic weight of some elements.

$$\text{Atomic Weight} = \text{Valency} \times \text{Equivalent weight.}$$

Initially, it was found that equivalent weight of Be is 4.5 and it is trivalent (V = 3), so the weight of Be was 13.5 and there is no space in Mendeleev's table for this element. So, after correction, it was found that Be is actually divalent (V = 2). So, the weight of Be became $2 \times 4.5 = 9$ and there was a space between Li and B for this element in Mendeleev's table.

Corrections were done in atomic weight of elements are – U, Be, In, Au, Pt.

(iv) Demerits of Mendeleev's periodic table –

(a) Position of hydrogen – Hydrogen resembles both, the alkali metals (IA) and the halogens (VIIA) in properties so Mendeleev could not decide where to place it.

(b) Position of isotopes – As atomic weight of isotopes differs, they should have been placed in different position in Mendeleev's periodic table. But there was no such place for isotopes in Mendeleev's table.

(c) Anomalous pairs of elements – There were some pair of elements which did not follow the increasing order of atomic wts.

eg : Ar and Co were placed before K and Ni respectively in the periodic table, but having higher atomic weights.

$$\underset{\text{in modified mendeleev P.T.}}{\begin{pmatrix} \text{Ar} & \text{K} \\ 39.9 & 39.1 \end{pmatrix}} \qquad \begin{pmatrix} \text{Te} & \text{I} \\ 127.5 & 127 \end{pmatrix}$$

$$\begin{pmatrix} \text{Co} & \text{Ni} \\ 58.9 & 58.6 \end{pmatrix} \qquad \begin{pmatrix} \text{Th} & \text{Pa} \\ 232 & 231 \end{pmatrix}$$

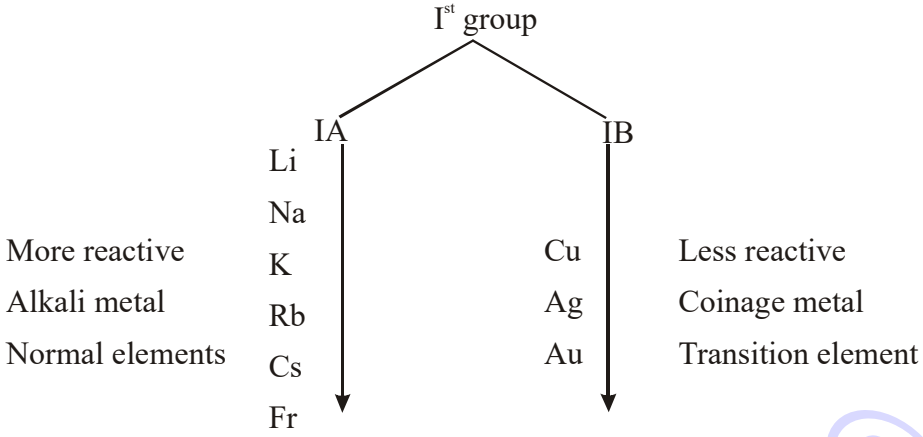
(d) Like elements were placed in different groups.

There were some elements like Platinum (Pt) and Gold (Au) which have similar properties but were placed in different groups in Mendeleev's table.

Pt	Au
VIII	IB

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(e) Unlike elements were placed in same group.



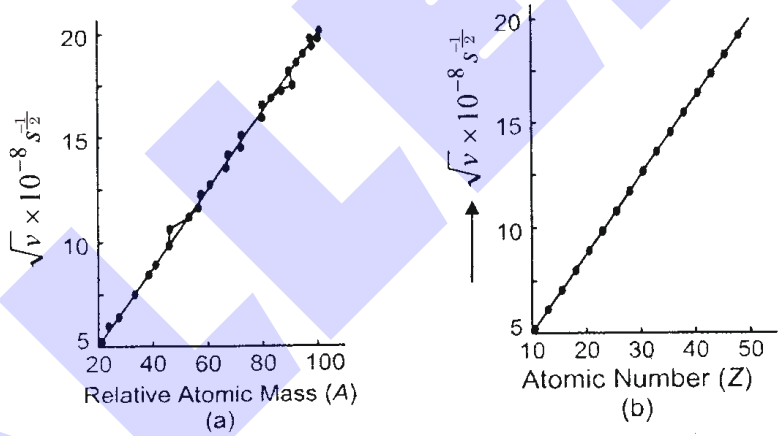
Cu, Ag and Au placed in Ist group along with Na, K etc. While they differ in their properties (Only similar in having ns^1 electronic configuration)

(F) MODERN PERIODIC TABLE (MODIFIED MENDELEEV PERIODIC TABLE)

(i) It was proposed by **Moseley (1913)**.

(ii) Modern periodic table is based on atomic number.

(iii) Moseley did an experiment in which he bombarded high speed electron on different metal surfaces and obtained X-rays.



Moseley's Experiment

He found out that $\sqrt{\nu} \propto Z$ (where ν = frequency) of X-rays from this experiment, Moseley concluded that the physical and chemical properties of the elements are periodic function of their atomic number. It means that when the elements are arranged in the increasing order of their atomic number, elements having similar properties gets repeated after a regular interval. This is also known as **'Modern Periodic Law'**.

(iv) Modern periodic law – The physical & chemical properties of elements are a periodic function of their atomic number.

(v) Characteristics of modern periodic table –

(a) 9 vertical columns called groups.

(b) IA to VIIA, IB to VIIB, VIII and 0

(c) Inert gases were introduced in periodic table by Ramsay.

(d) 7 horizontal series called periods.

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(G) LONG FORM / PRESENT FORM OF MODERN PERIODIC TABLE

(It is also called as 'Bohr-Burry & **Rang**, Werner Periodic Table.)

(i) It is based on the Bohr-Burry electronic configuration concept and atomic number.

(ii) This model was proposed by Rang & Werner

(iii) It consists of 7 horizontal periods and 18 vertical columns (groups)

(iv) According to I. U. P. A. C. 18 vertical columns are named as 1st to 18th group.

(v) The co-relation between the groups in long form of periodic table and in modern form of periodic table are given below.

IA, IIA, IIIB, IVB, VB, VIB, VIIB, VIII, IB, IIB, IIIA, IVA, VA, VIA, VIIA, 0

1 2 3 4 5 6 7 8, 9, 10 11 12 13 14 15 16 17 18

(vi) Elements belonging to same group have same no. of electrons in the outermost shell so their properties are similar.

Description of periods :

Period	n	Sub shell	No. of elements	Element	Name of Period
1.	1	1s	2	$_{1}H$ – $_{2}He$	Shortest
2.	2	2s, 2p	8	$_{3}Li$ – $_{10}Ne$	Short
3.	3	3s, 3p	8	$_{11}Na$ – $_{18}Ar$	Short
4.	4	4s, 3d, 4p	18	$_{19}K$ – $_{36}Kr$	Long
5.	5	5s, 4d, 5p	18	$_{37}Rb$ – $_{58}Xe$	Long
6.	6	6s, 4f, 5d, 6p	32	$_{55}Cs$ – $_{86}Rn$	Longest
7.	7	7s, 5f, 6d, 7p	32	$_{87}Fr$ – $_{118}Og$	Longest

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Extended or Long Form of the Periodic Table

s-Block Elements | p-Block Elements | d-Block Elements | f-Block Elements

Metals | Non metals | Metalloids

Group →	IA (1)	IIA (2)	IIIB (3)	IVB (4)	VB (5)	VIB (6)	VIIB (7)	(8)	VIII (9)	(10)	IB (11)	IIB (12)	IIIA (13)	IVA (14)	VA (15)	VIA (16)	VIIA (17)	0 (18)
Period ↓ 1	1 H 1.0079 Hydrogen																	2 He 4.0026 Helium
2	3 Li 6.940 Lithium	4 Be 9.0122 Beryllium											5 B 10.811 Boron	6 C 12.011 Carbon	7 N 14.007 Nitrogen	8 O 15.999 Oxygen	9 F 18.998 Fluonne	10 Ne 20.180 Neon
3	11 Na 22.990 Sodium	12 Mg 24.305 Magnesium											13 Al 26.982 Aluminium	14 Si 28.086 Silicon	15 P 30.974 Phosporus	16 S 32.066 Sulphur	17 Cl 35.453 Chlorine	18 Ar 39.948 Argon
4	19 K 39.098 Potassium	20 Ca 40.078 Calcium	21 Sc 44.956 Scandium	22 Ti 47.867 Titanium	23 V 50.941 Vanadium	24 Cr 51.996 Chrominum	25 Mn 54.938 Manganese	26 Fe 55.847 Iron	27 Co 58.933 Cobolt	28 Ni 58.693 Nickel	29 Cu 63.546 Copper	30 Zn 65.39 Zinc	31 Ga 69.723 Gallium	32 Ge 72.61 Germanium	33 As 74.922 Arsenic	34 Se 78.96 Selenium	35 Br 79.904 Bromine	36 Kr 83.80 Krypton
5	37 Rb 85.468 Rubidium	38 Sr 87.62 Strontium	39 Y 88.906 Yttrium	40 Zr 91.224 Zirconium	41 Nb 92.906 Niobium	42 Mo 95.94 Molybdenum	43 Tc 98 Technetium	44 Ru 101.07 Ruthenium	45 Rh 102.91 Rhodium	46 Pd 106.42 Palladium	47 Ag 107.87 Silver	48 Cd 112.41 Cadmium	49 In 114.82 Indium	50 Sn 118.71 Tin	51 Sb 121.76 Antimony	52 Te 127.60 Tellurium	53 I 126.90 Iodine	54 Xe 131.29 Xenon
6	55 Cs 132.91 Cesium	56 Ba 137.33 Barium	57 La* 138.91 Lanthanum	72 Hf 178.49 Hafnium	73 Ta 180.95 Tantalum	74 W 183.84 Tungsten	75 Re 186.21 Rhenium	76 Os 190.23 Osmium	77 Ir 192.22 Iridium	78 Pt 195.08 Platinum	79 Au 196.97 Gold	80 Hg 200.59 Mercury	81 Tl 204.38 Thallium	82 Pb 207.2 Lead	83 Bi 208.98 Bismuth	84 Po 210 Polonium	85 At 210 Astatine	86 Rn 222 Radon
7	87 Fr 223 Francium	88 Ra 226 Radium	89 Ac** 227 Actinium	104 Rf 267 Rutherfordium	105 Db 270 Dubnium	106 Sg 271 Seaborgium	107 Bh 270 Bohrium	108 Hs 277 Hassium	109 Mt 276 Meitnerium	110 Ds 281 Darmstadtium	111 Rg 282 Roentgenium	112 Cn 285 Copernicium	113 Nh 285 Nihonium	114 Fl 289 Flerovium	115 Mc 288 Moscovium	116 Lv 293 Livermorium	117 Ts 294 Tennessine	118 Og 294 Oganesson

6	*Lanthanide Series	58 Ce 140.12 Cerium	59 Pr 140.91 Praseodymium	60 Nd 144.24 Neodymium	61 Pm 145 Promethium	62 Sm 150.36 Samarium	63 Eu 151.96 Europium	64 Gd 157.25 Gadolinium	65 Tb 158.93 Terbium	66 Dy 162.50 Dyspro-	67 Ho 164.93 Holmium	68 Er 167.26 Erbium	69 Tm 168.93 Thulium	70 Yb 173.07 Ytterbium	71 Lu 174.97 Lutetium
7	**Actinide Series	90 Th 232.04 Thorium	91 Pa 231.04 Protactinium	92 U 238.03 Uranium	93 Np 237 Neptunium	94 Pu 244 Plutonium	95 Am 243 Americium	96 Cm 247 Curium	97 Bk 247 Berkelium	98 Cf 251 Californium	99 Es 252 Einsteinium	100 Fm 257 Fermium	101 Md 258 Mendelevium	102 No 259 Nobelium	103 Lr 262 Lawrencium

IUPAC designations of groups of elements are given in brackets

CLASSIFICATION OF ELEMENTS INTO s, p, d & f BLOCK ELEMENTS:

s – block :

(i) configuration $n s^{1-2}$ **(ii)** last e^- enters in s orbital

(iii) two groups I A or 1 ; II A or 2

p – block :

(i) configuration $n s^2 n p^{1-6}$ **(ii)** last e^- enters in p orbital

(iii) six groups III A , IV A , V A , VI A , VII A , zero or 13 , 14 , 15 , 16 , 17 , 18

d – block : [Transition Elements]

(i) configuration $(n-1) d^{1-10} n s^{0-2}$ **(ii)** last e^- enters in d orbital

(iii) their two outermost shell are incomplete

(iv) 10 groups III B , IV B , V B , VI B , VII B , VIII (Triad) , I B , II B or 3 , 4 , 5 , 6 , 7 , (8 , 9 , 10) , 11 , 12 .

(v) four series 3 d , 4 d , 5 d , 6 d belong to 4th, 5th, 6th & 7th period respectively in long form of Periodic table.

f – block : [Inner Transition]

(i) configuration $(n-2) f^{0-14} (n-1) d^{0-2} ns^2$

(ii) last e^- enters in f orbital

(iii) two series 4 f Lanthanides & 5 f Actinides belong to 6th & 7th period respectively in long form of Periodic table.

Neil Bohr's classification of elements :

Using electronic configuration as the criteria, elements are of four types. The classification of the elements into these groups is dependent on the extent to which the s, p, d and f orbitals are filled.

Inert Gases :

(a) s – and p-orbitals of the outer most shell of these elements are completely filled. The outermost electronic configuration is ns^2np^6.

(b) Helium is also inert gas but its electronic configuration is $1s^2$

Representative or Normal Elements :

(a) Outermost shell of these elements is incomplete. The number of electrons in the outermost shell is less than eight.

(b) s-and p-block elements except inert gases are called normal or representative elements.

Transition Elements :

(a) These metals were placed between s-block metals and p-block elements so, are named transition metals.

(b) Their outermost electronic configuration is similar to d-block elements i.e. $(n-1) d^{1-10} ns^{1-2}$.

(c) Last two shells of these elements namely outermost and penultimate shells are incomplete.

(d) The last shell contains one or two electrons and the penultimate shell may contain more than eight and up to eighteen electrons.

(e) According to definition of transition elements, those elements which have partly filled d-orbitals in neutral state or in any stable oxidation state are called transition elements. According to this definition Zn, Cd and Hg (IIB group) are d-block elements but not transition elements because these elements have d^{10} configuration in neutral as well as in stable +2 oxidation state.

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Inner Transition Elements :

(a) In these elements last three shells i.e. last, penultimate and prepenultimate shells are incomplete.

(b) These are related to IIIB i.e. group 3.

(c) The last shell contains two electrons. Penultimate shell may contain eight or nine electrons and pre-penultimate shell contains more than 18, up to 32 electrons.

(d) Their outemost electronic configuration is similar to f-block element
i.e. $(n-2)\,f^{0-14}\,(n-1)\,d^{0-2}\,ns^2$

METALS, NON-METALS & METALLOIDS

Apart from classifying elements into s, p, d and *f*-blocks, there is yet another broad classification of elements based on their properties. The elements can be broadly classified into

(a) Metals :

Majority of the elements in periodic table are metals and appears on the left side of the periodic table.

Properties :

(i) These are usually solid at room temperature [exception - mercury]

(ii) They have high melting and boiling point [exception Gallium & Cesium have very low melting point (303 K and 302 K respectively)]

(iii) They are good conductor of heat and electricity.

(iv) They are malleable (can be flattened into thin sheets by hammering) and ductile (can be drawn into wires)

(b) Non-Metals :

These are placed at the top right hand side of periodic table. As we move horizontally along a period, the property of elements changes from metallic (on left) to non-metallic (on the right).

Properties :

(i) These are usually solids or gases at room temperature.

(ii) They have low melting point and boiling point (exception : Boron, Carbon).

(iii) Most Non-metallic solids are brittle and are neither malleable nor ductile.

(c) Metalloids (Semi-metals):

Properties of these elements show the characteristics of both metals and non-metals. Silicon (Si), Germanium(Ge), Arsenic(As), Antimony(Sb) and Tellurium(Te) are metalloids.

ESTIMATING POSITION OF AN ELEMENT FROM ITS ELECTRONIC CONFIGURATION

The last electron enters in which subshell gives idea of its block.

[☺ ***Think*** :$1s^1$ and $1s^2$ belongs to which block]

Period number = Principal quantum number of valence shell electron in ground state electronic configuration.

Group number for s block = number of valence shell electrons

Group number for p block = 10 + number of valence shell electrons

Group number for d block = number of [ns + (n–1) d] electrons

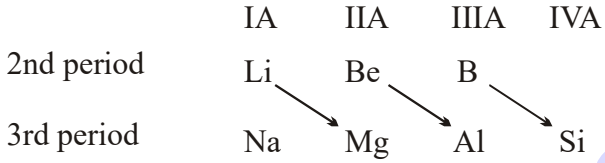
Group number for f-block = 3

[☺ Use these carefully while locating the position.]

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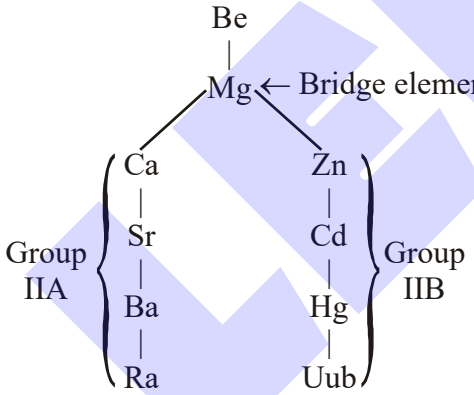
SOME COMMONLY USED TERMS

1. ***Noble Gases :*** Element of group 18 are called noble gases. These are also called as inert gases because their outermost ns and np orbitals are completely filled (except He and $1s^2$) and these gases are non-reactive in nature under ordinary conditions.
2. ***Typical elements :*** Elements of second and third period are known as typical elements.
3. ***Diagonal relationship :*** Properties of elements of second period resemble with the element of third period. This resemblance between properties of 2nd & 3rd period is called diagonal relationship.



4. ***Bridge elements :***

The typical elements of third period are also called bridge elements as the division between two subgroups A and B starts from these elements. In second group Mg acts as a bridge element. The properties of bridge element are some what mixed of the elements of two subgroups as magnesium shows similarities with alkaline earth metals (IIA) on one hand and with zinc metals (IIB) on the other.



IUPAC NOMENCLATURE OF THE ELEMENT :

The names are derived by using roots for the three digits in the atomic number of the element and adding the ending –ium. The roots for the number are

Digit	Name	Abbreviation
0	nil	n
1	un	u
2	bi	b
3	tri	t
4	quad	q
5	pent	p
6	hex	h
7	sept	s
8	oct	o
9	enn	e

Thus element with atomic number 109 will be named as **une** (**u** for 1, **n** for 0 and **e** for 9). Table summarises the names of the elements with atomic number above 100.

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PART-B (PERIODIC PROPERTIES)

PERIODICITY :

(a) The regular gradation in properties from top to bottom in a group and from left to right in a period is called periodicity in properties.

(b) In a period, the ultimate shell remain same, but the number of electrons gradually increases.

(c) In a group, the number of electrons in the ultimate shell remains same, but the values of n increases.

Cause of periodicity

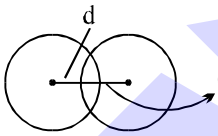
(a) The cause of periodicity in properties is due to the same outermost shell electronic configuration repeating over regular intervals.

(b) In the periodic table, elements with similar properties occur at intervals of 2, 8, 8, 18, 18 and 32. These numbers are called magic numbers.

ATOMIC RADIUS :

Since there is a problem in calculating actual size of atom, three types of radii can be defined :

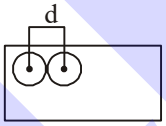
Type-I ***Covalent radius :***



$C_r = \frac{d}{2}$

[Used for H_2, Cl_2 and such molecules]

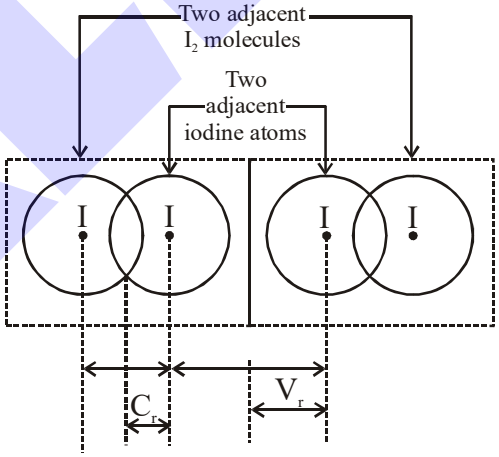
Type-II ***Metallic Radius :***



$M_r = \frac{d}{2}$

[Used for metals]

Type-III **VanderWaal's Radius or Collision radius**



VanderWaal's radius = $\frac{1}{2}\times$ Internuclear distance between nuclei of two neighbouring atoms belonging to nearest molecules.

VanderWaal's radius > Metallic radius > Covalent radius

The VanderWaal's radius and covalent radius of chlorine atom are 1.80 Å and 0.99 Å respectively.

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Ionic Radius

A neutral atom changes to a cation by the loss of one or more electrons and to an anion by the gain of one or more electrons. The magnitude of charge on cation and anion is equal to the number of electrons lost or gained respectively. The ionic radii of the ions present in an ionic crystal may be calculated from the internuclear distance between the two ions.

(a) ***Radius of Cation***

Radius of cation is smaller than that of corresponding atom. Since due to removal of electron(s), Z_{eff} increases.

(b) ***Radius of an Anion***

Radius of an anion is invariably larger than that of the corresponding atom, since due to addition of electron(s) Z_{eff} decreases.

Factors affecting atomic radius:

(a) Z_{eff} increases, atomic radius decreases — $Li > Be > B > C > N > O > F$

(b) Number of shell(n) increases, atomic radius increases — $Li < Na < K < Rb < Cs$

(c) Screening effect increases, atomic radius increases.

(d) Magnitude of –ve charge increases, atomic radius increases — $O < O^{-} < O^{-2}$

(e) Magnitude of +ve charge increases, atomic radius decreases — $Mn > Mn^{+2} > Mn^{+3} > Mn^{+4}$

(f) Bond order increases, atomic radius decreases — $>N-N< > -N=N- > N\equiv N$

Periodic Trend :

(a) ***For normal elements:***

(i) ***Across a period :*** It decreases from left to right in a period as $Z_{eff.}$ increases.

Ex. Ne > Li > Be > B > C > N > O > F

(ii) ***In a group :*** It increases from top to bottom in a group as number of shells increases.

Ex. Li < Na < K < Rb < Cs

☞ ***Note :*** In III A group size of Al and Ga is nearly same (transition contraction)

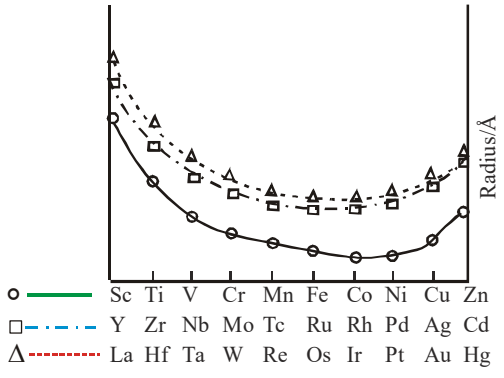
(b) ***For inert gases:***

In respective period generally, the atomic radius of inert gas is largest, because for inert gas VanderWaal's radius is defined. The VanderWaal's radius of inert gases also increases on moving from top to bottom in the group.

(c) ***For transition elements:***

From left to right in a period:

Element	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Covalent radius (Å)	1.44	1.32	1.22	1.17	1.17	1.17	1.16	1.15	1.17	1.25



Trends in atomic radii of transition elements

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In a group :

(i) The atomic radius of elements increases on moving down the first transition series (3d) to second transition series (4d). This is due to the increases in number of shells with the increase in atomic number.

(ii) The atomic radii of second (4d) and third (5d) transition series in a group is almost same except Y(39) and La(57)

(d) ***For inner transition elements :***

As we move along the lanthanide series, there is a decrease in atomic as well as ionic radius. The decrease in size is regular in ions but not so regular in atoms. This is called lanthanide contraction*.

Exceptions :

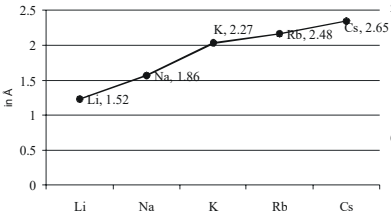
(1) Noble gases have largest atomic sizes [Vander waal radii]. However, their covalent radii are smaller *e.g.* Xe.

(2) Size of Al > Ga, [Z_{eff} increasing]

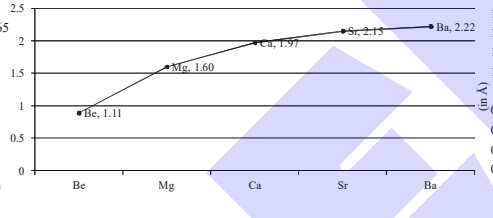
(3) Size of Hf & Zr are same (lanthanide contraction)

Graphical representation of atomic radius :

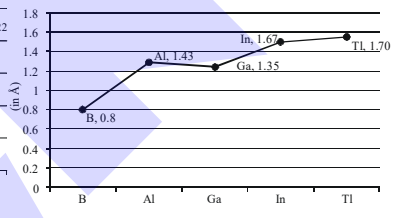
ALKALI METALS



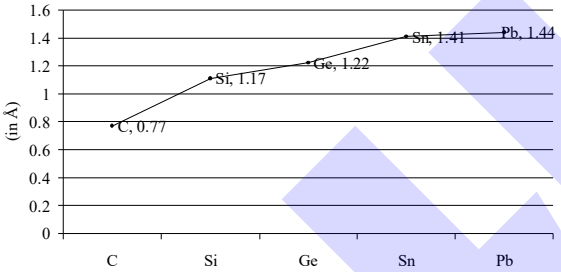
ALKALINE EARTH METALS



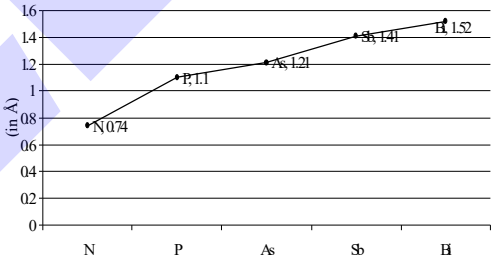
BORON FAMILY



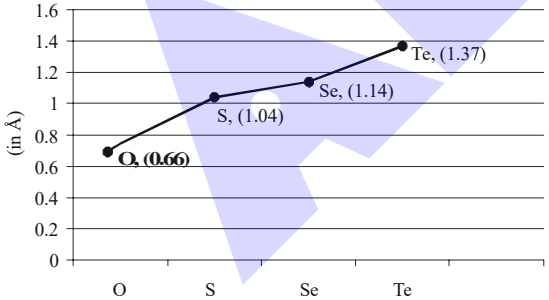
CARBON FAMILY



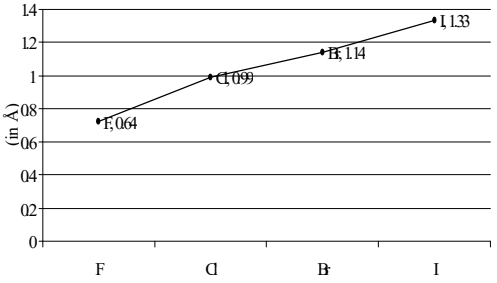
NITROGEN FAMILY (PNICOGENS)



CHALCOGENS



HALOGENS



[☺ what can you predict or say about the increment in size along a group and decrement along a period]

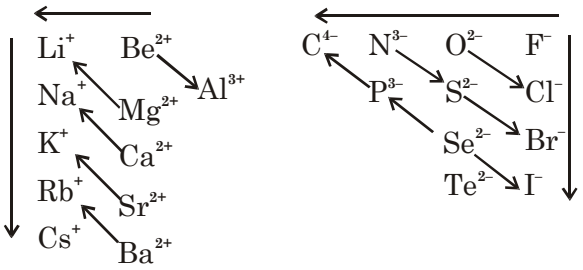
ISOELECTRONIC MONOATOMIC SPECIES [Size ∝ 1/Z] :

(i) S^{2-} , Cl^{-} , K^{+} , Ca^{2+} , Sc^{3+} **(ii)** H^{-} , He , Li^{+}

(iii) O^{-2} , F^{-} , Na^{+} , Mg^{+2}, Al^{+3} **(iv)** $\frac{\text{radius of cation}}{\text{radius of anion}} = \frac{Z_{eff} \text{ of Anion}}{Z_{eff} \text{ of Cation}}$

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Note :- In the direction of arrow (→) ionic size increases.



IONISATION ENERGY :

Amount of energy required to remove the most loosely bonded electron from an isolated gaseous atom from its ground state electronic configuration.

Units : kJ mol^{-1}, k cal mol^{-1}, eV per atom.

Ionisation is endothermic (endoergic) i.e. requires energy. Hence $\Delta H_{ionsation}$ is +ve.

$M + \text{Energy } (IE_1) \longrightarrow M^+ + e^-$

$M \longrightarrow M^+ + e^- \quad \Delta H = IE_1$

$M^+ \longrightarrow M^{+2} + e^- \quad \Delta H = IE_2$

$M^{+2} \longrightarrow M^{+3} + e^- \quad \Delta H = IE_3$

] Successive ionisation energy

$IE_3 > IE_2 > IE_1$ (always)

FACTORS AFFECTING IONISATION ENERGY :

(1) Atomic size : Varies inversely

(2) Screening effect : Varies inversely

(3) Nuclear charge (Z) : Varies directly

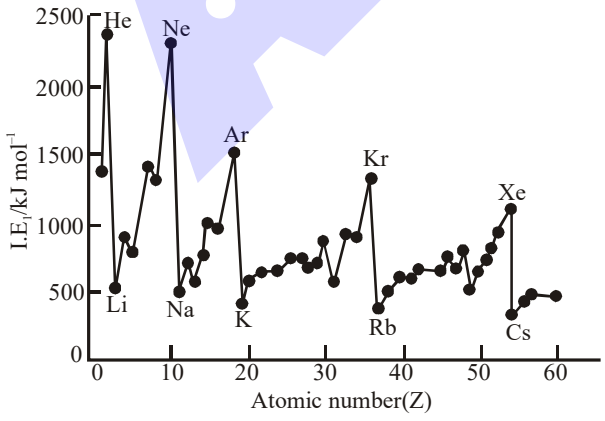
(4) Special electronic configuration of outermost electron (half filled / fully filled)

(5) Type of orbitals involved in Ionisation : s > p > d > f.

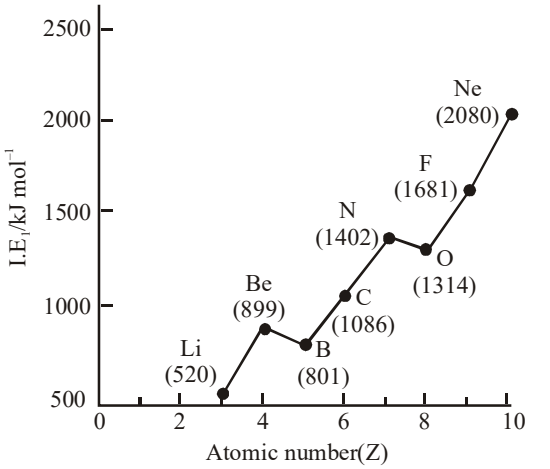
Note : Half filled and full filled inner orbitals, affects d-block and *f*-block trends.

General Trend: Along period I.E. increases [with some exception] [Z_{eff} ↑]

Along a group I.E. decrease [Z_{eff} constant, n ↑]



Variation of first ionization energy ($I.E_1$) with atomic number for elements with Z = 1 to 60



First ionization energy ($I.E_1$) of elements of the second period as a function of atomic number(Z)

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Exception :

(1) Along a period, half filled and fully filled configurations have higher I.E.
e.g. Be > B and N > O.

(2) along a group, Ga ≃ Al

PROPERTIES AFFECTED BY IONISATION ENERGY :

(1) Metallic character (Varies inversely)

(2) Reducing power (Varies inversely)

(3) Tendency to stay in which state A^{+1}. A^{+2} or A^{+3}

☞ ***Note :***

(a) Helium (He) has the highest ionisation energy (IE_1) among all the elements and Caesium (Cs) has the least (IE_1) value.

(b) Ionisation potential of inert gases is very high due to most stable s^2p^6 electronic configuration.

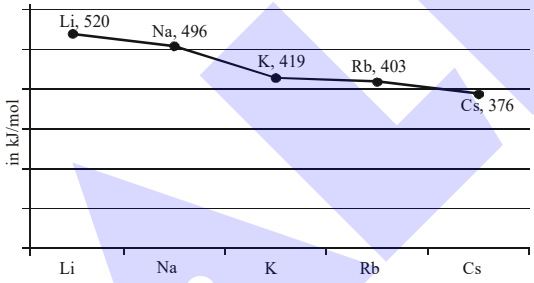
Element	He	Ne	Ar	Kr	Xe	Rn
IE_1(eV)	24.5	21.6	15.8	14.0	12.1	10.7

(c) For isoelectronic species I.E. increases with positive charge and decreases with negative charge.

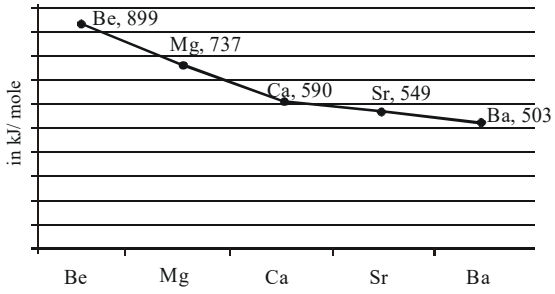
e.g. $Al^{+3} > Mg^{+2} > Na^{+} > F^{-} > O^{-2} > N^{-3}$

Graphical representation of ionisation energy :

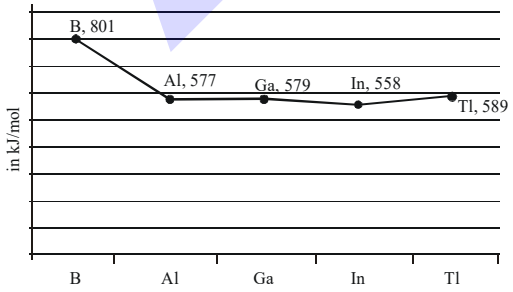
ALKALI METALS



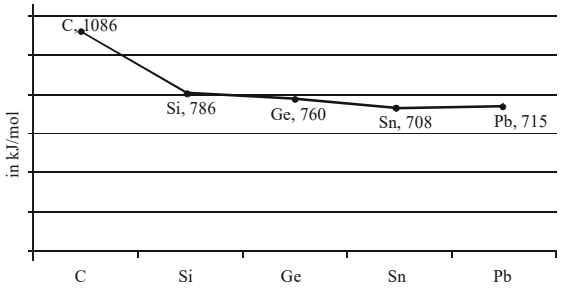
ALKALINE EARTH METALS



BORON FAMILY

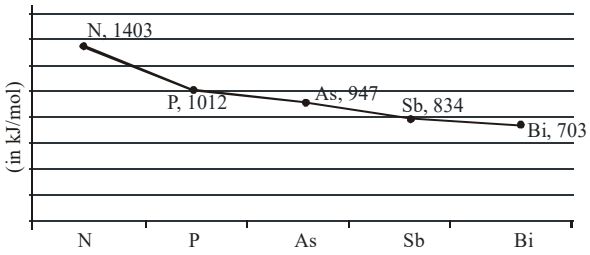


CARBON FAMILY

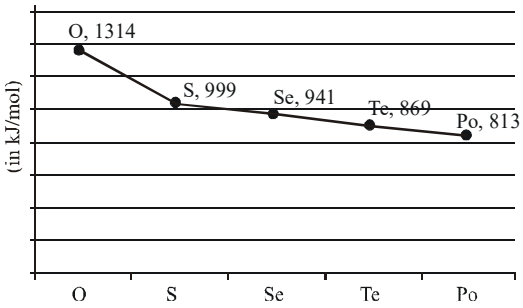


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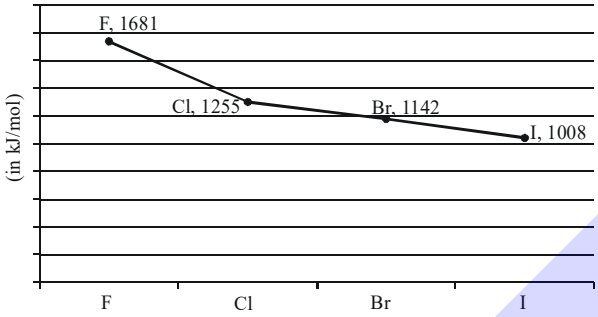
NITROGEN FAMILY (PNICOGENS)



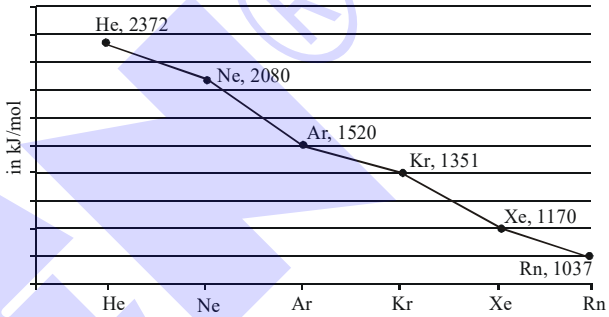
CHALCOGENS



HALOGENS



NOBLE GASES



Ionisation energy of d-block elements :

1st, 2nd, 3rd IE's are increasing from left to right for 1st Transition series, but not regularly.

For 2nd IE Cr > Fe > Mn and Cu > Zn

For 3rd IE Mn > Cr > Fe and Zn has highest.

Element		Sc	Ti	v	Cr	Mn	Fe	Co	Ni	Cu	Zn
Atomic number		21	22	23	24	25	26	27	28	29	30
Electronic configuration											
	M	$3d^14s^2$	$3d^24s^2$	$3d^34s^2$	$3d^54s^1$	$3d^54s^2$	$3d^64s^2$	$3d^74s^2$	$3d^{10}4s^2$	$3d^{10}4s^1$	$3d^{10}4s^2$
	M^{2+}	$3d^1$	$3d^2$	$3d^3$	$3d^4$	$3d^5$	$3d^6$	$3d^7$	$3d^8$	$3d^9$	$3d^{10}$
	M^{3+}	[Ar]	$3d^1$	$3d^2$	$3d^3$	$3d^4$	$3d^5$	$3d^6$	$3d^7$	–	–
Enthalpy of atomisation, $\Delta_aH^\ominus$/kJ mol^{-1}		326	473	515	397	281	416	425	430	339	126
Ionisation Enthalpy, $\Delta_iH^\ominus$/kJ mol^{-1}											
$\Delta_iH^\ominus$	I	631	656	650	653	717	762	758	736	745	906
	II	1235	1309	1414	1592	1509	1561	1644	1752	1958	1734
	III	2393	2657	2833	2990	3260	2962	3243	3402	3556	3829

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ELECTRON AFFINITY & ELECTRON GAIN ENTHALPY (EGE) :

Electron Affinity : Amount of energy released when an electron is added to an isolated gaseous atom.
Units : k J mol^{-1}, k Cal mol^{-1} and eV per atom.
Addition of electron results in release of energy in most of the cases but for addition of second electron energy is always required. The sum of EGE_1 & EGE_2 is +ve (energy required)

$E\ A\ \alpha\ \frac{1}{\text{atomic size}}\ \alpha\ Z_{eff}$ (Cl has the highest E.A.)

Electron gain Enthalpy (EGE) : When expressed in terms of enthalpy change (ΔH_{eg}) then it is termed as EGE Remember that ΔH = – ve for exothermic change.

For EA_1, energy is released ∴ ΔH_{eg1} = – ve (Not always)
For EA_2, EA_3 .....energy is required ∴ ΔH_{eg2} = + ve (always)
In general $EA_1 + EA_2$, energy is required. ∴ $(\Delta H_{eg1} + \Delta H_{eg2}) > 0$

Note : $\Delta H_{e.g.} = -EA + \frac{5}{2}RT$

$M_{(g)} \underset{\text{E.A. of } M^+_{(g)}}{\overset{\text{I.E. of } M_{(g)}}{\rightleftharpoons}} M^+_{(g)}$

Ionisation energy of element is equal to electron affinity of it's cation.

FACTORS AFFECTING ELECTRON AFFINITY :

(1) ***Atomic size :*** Varies inversely
(2) ***Nuclear charge :*** Varies directly
(3) For stable electronic configuration i.e. half filled and fully filled shells EA decreases.

General Trend : Along a period, electron affinity increases [with a few exceptions] as Z_{eff} ↑.
Along a group, electron affinity decreases after 3rd period. Between 2nd and 3rd period in p block electron affinity of 2nd period is lesser.

Exception :

(1) A fully filled and half filled configuration have low values of EA or even sometimes energy is required rather than getting released.
(2) 2nd period has lower value than 3rd owing to repulsion between electrons in small sized 2nd period elements.

Electron affinity of the main-group elements (in electron volts)

1							18
H							**He**
+0.754	2	13	14	15	16	17	–0.5
Li	**Be**	**B**	**C**	**N**	**O**	**F**	**Ne**
+0.618	≤ 0	+0.277	+1.263	–0.07	+1.461	+3.399	–1.2
Na	Mg	Al	Si	P	S	Cl	Ar
+0.548	≤ 0	+0.441	+1.385	+0.747	+2.077	+3.617	–1.0
K	**Ca**	**Ga**	**Ge**	**As**	**Se**	**Br**	**Kr**
+0.502	-	+0.03	+1.2	+0.81	+2.021	+3.365	–1.0
Rb	**Sr**	**In**	**Sn**	**Sb**	**Te**	**I**	**Xe**
+0.486	-	+0.3	+1.2	+1.07	+1.971	+3.059	–0.8

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ELECTRO NEGATIVITY :

Property of an atom in a molecule

F has highest electronegativity in the periodic table.

Decreasing order → F > O > Cl ≃ N > Br > S ≃ C > I > H.

Pauling Scale : $|X_A - X_B| = 0.208\sqrt{\Delta_{A-B}}$ Δ in kcal/mol

$|X_A - X_B| = 0.102\sqrt{\Delta_{A-B}}$ Δ in kJ/mol

$\Delta_{A-B} = E_{A-B} - (E_{A-A} \times E_{B-B})^{1/2}$

Where, E_{A-B} = Bond energy of A–B; E_{A-A} = Bond energy of A–A; E_{B-B} = Bond energy of B–B

Mulliken's Scale : $X_M = \frac{IP + EA}{2}$ (IP & EA are expressed in eV)

Mulliken's values of E N are about 2.8 times higher than values on Pauling's scale.

i.e. $X_P = \frac{X_M}{2.8}$

Allred–Rochow's Scale : $X_{AR} = \frac{0.359Z_{eff}}{r^2}$; $X_P = X_{AR} + 0.744$

r = covalent radius of atom (in Å)

Z_{eff} = Effective nuclear charge on periphery

FACTORS AFFECTING ELECTRO NEGATIVITY :

(1) ***Nuclear attraction :*** Varies directly

(2) ***Atomic radius :*** Varies inversely

(3) ***Charge on ions :*** More positive charge, more electronegativity and more –ve charge, less electronegativity.

(4) ***Hybridisation :*** To be discussed later in chemical bonding chapter.

General Trends : Along a period, electronegativity increases

Along a group, electronegativity decreases

Exceptions : Not Noteworthy.

☞ **Note: Electronegativity of F > Cl but Electron affinity of Cl > F**

Electronegativity of Some Elements (on Paulling's Scale)

H 2.1						
Li 1.0	Be 1.5	B 2.0	C 2.5	N 3.0	O 3.5	F 4.0
Na 0.9	Mg 1.2	Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0
K 0.8	Ca 1.0	Ga 1.6	Ge 1.8	As 2.0	Se 2.4	Br 2.8
Rb 0.8	Sr 1.0	In 1.7	Sn 1.8	Sb 1.9	Te 2.1	I 2.5
Cs 0.7	Ba 0.9	Tl 1.8	Pb 1.8	Bi 1.9	Po 2.0	At 2.2
Fr 0.7	Ra 0.9					

In **Pauling's Scale,** elements having almost same electronegativity are-

C ≃ S ≃ I ≃ 2.5
N = Cl = 3.0
P = H = 2.1
Cs= Fr = 0.7
Be= Al = 1.5

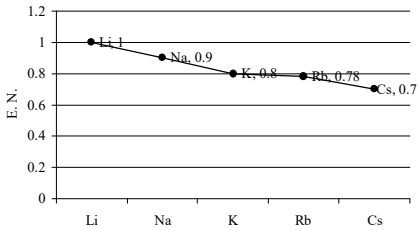
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PROPERTIES DEPENDENT ON ELECTRO NEGATIVITY :

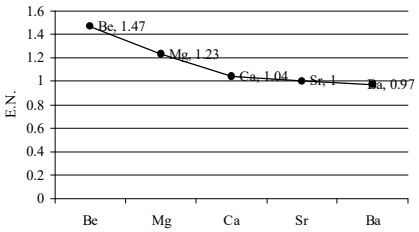
(1) % ionic character (2) Strength of bond
(3) Bond Length (4) Nature of hydrides
(5) Nature of hydroxide.

Graphical representation of Electronegativity :

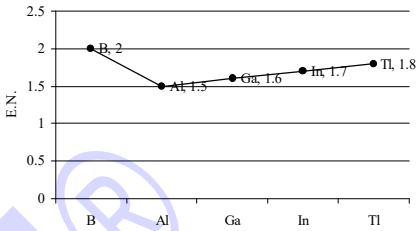
ALKALI METALS



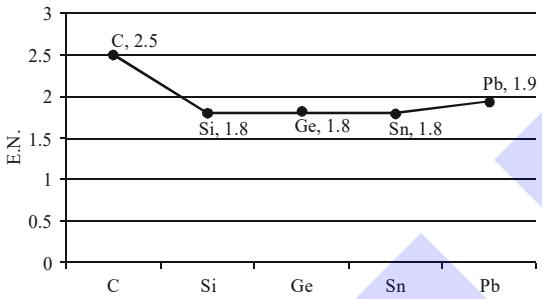
ALKALINE EARTH METALS



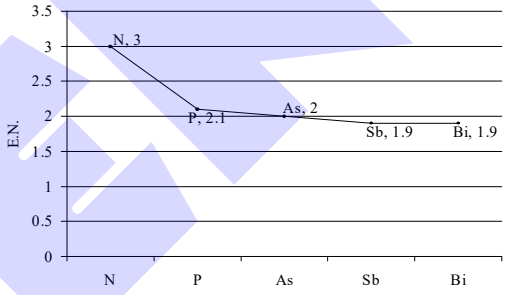
BORON FAMILY



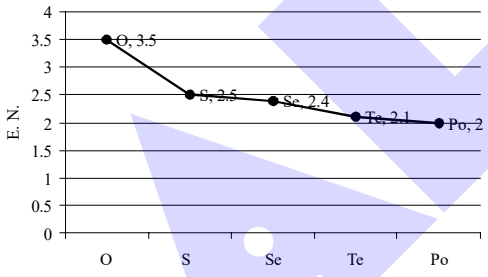
CARBON FAMILY



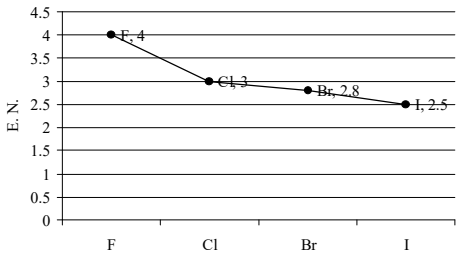
NITROGEN FAMILY (PNICOGENS)



CHALCOGENS



HALOGENS



NATURE OF OXIDES

Amphoteric Oxides

M_2O	MO	M_2O_3	MO_2	M_2O_5	Amphoteric/Neutral
Ga_2O	BeO	Al_2O_3	MnO_2	V_2O_5	H_2O
	ZnO	Cr_2O_3	PbO_2		
	SnO	Ga_2O_3	SnO_2		
	PbO	As_2O_3			
		Sb_2O_3			

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LATTICE ENERGY :

The amount of energy released during the formation of 1 mole crystal lattice from consituent gaseous ions.

$Na^+(g) + Cl^-(g) \rightarrow NaCl(s)$ Lattice energy = –788 kJ mol^{-1}

OR

The lattice energy of an ionic compound is the energy required to separate 1 mole of solid ionic substance completely into gaseous ions.

$NaCl(s) \rightarrow Na^+(g) + Cl^-(g)$ Lattice energy = +788 kJ mol^{-1}

Lattice energies are large and positive because of attraction between positive and negative ions. The potential energy of two interacting charged particles is given by:

$$E = K\frac{Q_1Q_2}{r}$$ where, $r = r^+ + r^-$; $r^+ \rightarrow$ radius of cation, $r^- \rightarrow$ radius of anion

Where Q_1 and Q_2 are the charge on the particles in coulombs, and r is the distance between their centres in meters. The constant K has the value 9.0×10^9 J – m/C^2.

Factors affecting lattice energy :

(a) The lattice energy increases as the charge on the ions increases and as their radii decreases.

(b) The magnitude of lattice energies however depends primarily on the ionic charges because ionic radii do not vary over a wide range.

For Example :

The lattice energy order for some compounds is given below.

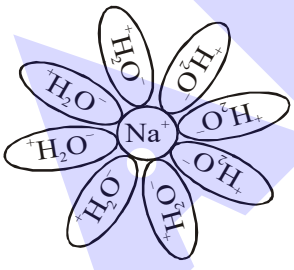
LiI < LiBr < LiCl < LiF (Due to small size of halide ion F^- alongwith small Li^+)

HYDRATION ENERGY :

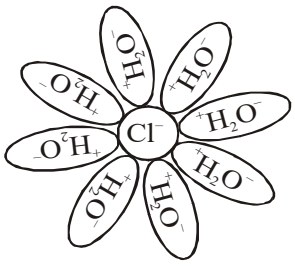
It is the energy released when 1 mol of gaseous ions are hydrated in water. It is directly proportional to nuclear charge and inversely proportional to size. It always decreases down the group.

(a) As the charge density of ion increases hydrated size (or aqueous radius) increases.

Size : $Li^+(aq) > Na^+(aq) > K^+(aq) > Rb^+(aq) > Cs^+(aq)$



Hydration of Na^+



Hydration of Cl^-

(b) As the hydrated size of ion increases ionic mobility decreases, which thus, decreases conductivity of ions.

Mobility : $Li^+(aq) < Na^+(aq) < K^+(aq) < Rb^+(aq) < Cs^+(aq)$

Conductivity : $Li^+(aq) < Na^+(aq) < K^+(aq) < Rb^+(aq) < Cs^+(aq)$

(c) Hydration energy also affects the solubility of ionic compounds. If hydration energy is greater than lattice energy then ionic compound will be soluble in water. More is the hydration energy, greater is the solubility, whereas, if lattice energy decreases, solubility of ionic compound increases.

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BORN-HABER CYCLE :

The Born-Haber cycle is an approach to analyze reaction energies.

Various energy terms

⇒ **Heat of sublimation ($\Delta H_{sub.}$) :** It is heat required to change one mole of a substance from solid state to gaseous state.

$A_{(s)} \longrightarrow A_{(g)}$; $\Delta H_{sub.}$ = +ive

⇒ **Heat of vaporisation ($\Delta H_{vap.}$) :** It is heat required to change one mole of liquid substance into gas.

$A_{(\ell)} \longrightarrow A_{(g)}$; $\Delta H_{vap.}$ = +ive

⇒ **Heat of fusion (ΔH_{fus}) :** It is heat required to change one mole of solid substance into liquid.

$A_{(s)} \longrightarrow A_{(\ell)}$; $\Delta H_{fus.}$ = +ive

⇒ **Heat of atomization (ΔH_{atm}) :** It is change in enthalpy when one mole of a substance is converted into its constituent gaseous atoms.

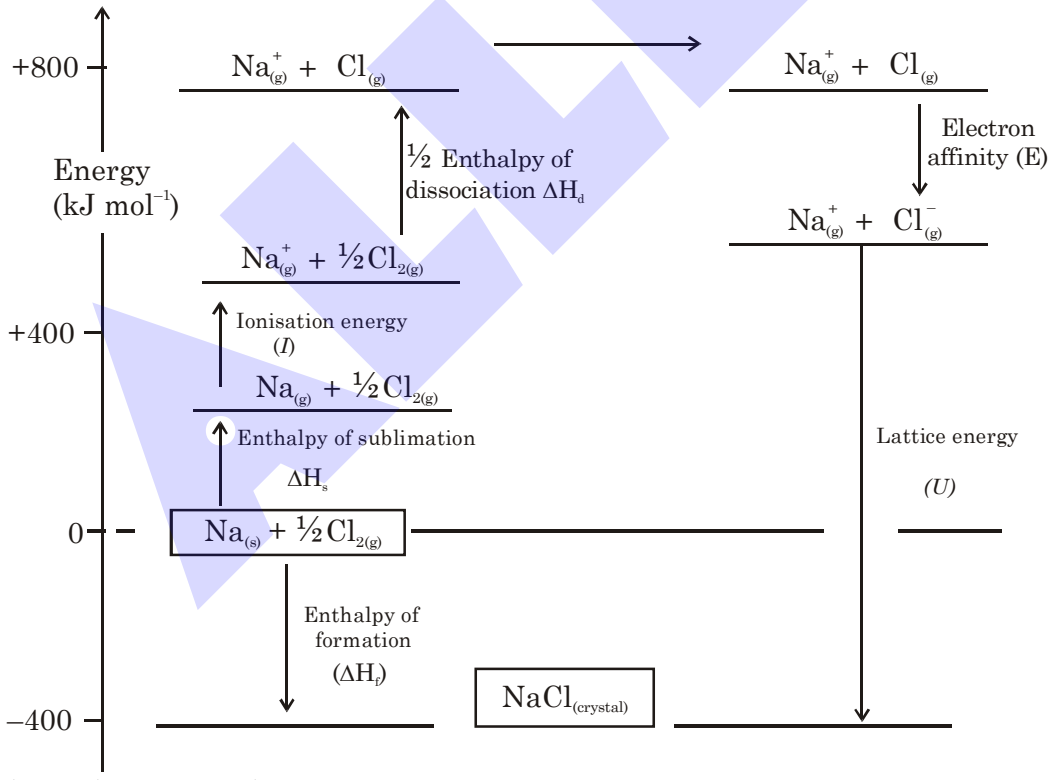
⇒ **Heat of solution ($\Delta H_{sol.}$) :** It is the change in enthlapy when one mole of a solute is dissolved in excess amount of solvent.

$NaCl_{(S)} + aq. \longrightarrow Na^+_{(aq.)} + Cl^-_{(aq.)}$, $\Delta H_{sol.\ of\ NaCl(s)}$

⇒ **Bond dissociation energy (B.D.E.) and Bond Energy (B.E.) :** It is the amount of heat require to break one mole of bond between two atoms of a gaseous molecule into gaseous atoms.

⇒ **Heat of formation (ΔH_f) :** It is the change in enthalpy when one mole of substance is formed from it's element in standard state.

Born -Haber cycle for $NaCl_{(s)}$



$\Delta H_f = \Delta H_S + I + \frac{1}{2}\Delta H_d + E + U$

Examples : MgO, $A\ell F_3$

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MISCELLANEOUS CHEMICAL PROPERTIES :

1. Periodicity of hydra acids :

(a) Acidic character of hydra acid increases from left to right in a period.

(b) Acidic character of hydra acid increases from top to bottom in a group.

2. Periodicity of oxy acids :

(a) Acidic character of oxy acid increases from left to right in a period.

(b) Acidic character of oxy acid decreases from top to bottom in a group.

3. Periodicity of nature of oxide :

(a) On moving from left to right in a period acidic nature of oxide generally increases.

e.g. $CO_2 < P_2O_5 < SO_3 < Cl_2O_7$

(b) On moving from top to bottom in a group acidic nature of oxide generally decreases.

4. General trends

(a) ***Hydration energy*** decreases along a group.

(b) ***Lattice energy*** decreases along a group.

:: *Some points to Remember* ::

(without considering radioactive elements)

1. Second most electronegative element is Oxygen

2. Hydrogen is the lightest element and Lithium is lightest metal.

3. Helium has the highest value of I.P.

4. In periodic table metalloids are only in p-block.

5. Total gaseous elements are 11 (He, Ne, Ar, Kr, Xe, Rn, H_2, N_2, O_2, Cl_2, F_2)

6. Liquid metal at room temperature is – Hg.

7. Diamond is hardest natural substance.

8. Halogens have highest electron affinity and amongst them, Cl has the highest amongst them.

9. The largest cation of the periodic table = Cs^+.

10. The smallest cation of the periodic table = H^+.

11. The smallest anion of the periodic table = F^-.

12. The biggest element of periodic table = Fr.

13. The smallest element of periodic table = H.

14. Br is liquid non-metal element at room temperature.

15. Osmium is the heaviest element known.

16. Fluorine is the most electronegative element.

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